PROJECT REPORT

SOCIAL MEDIA CONTENT ENGAGEMENT ANALYSIS

Team ID: DA-020
University: Invertis University
Department: B.Tech CSE, 3rd Year
Mentor: Mr. Arsh Singh
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Abstract

This project uses Python-based data analysis to explore social media post engagement across platforms, content formats, and publishing times. The notebook is designed to clean and prepare post-level data, calculate total engagement and an impressions-based engagement rate, and present descriptive comparisons through static and interactive visualizations. The report describes the implemented methodology. Dataset-specific numerical findings are not stated because the CSV data was not included with the uploaded notebook.

1. Introduction

Social media posts can receive different levels of interaction depending on their format, platform, and publishing time. Data analysis can help organize these interactions and describe patterns in a dataset. This project examines likes, comments, shares, impressions, reach, post type, platform, and timestamp fields where available.

The work is descriptive: it summarizes the supplied records and does not establish that a particular posting time or content format causes higher engagement.

2. Project Objectives

• Inspect and clean social media post data.

• Calculate total engagement from likes, comments, and shares.

• Calculate an impressions-based engagement rate.

• Compare engagement by post type and platform.

• Explore posting-hour and monthly engagement patterns.

• Present findings through static and interactive visualizations.

3. Dataset Description

The notebook expects a CSV file named Social_Media_Engagement.csv. The README identifies the expected location as dataset/Social_Media_Engagement.csv and notes that Google Colab can prompt the user to upload the CSV if it is not found.

The notebook uses fields including likes, comments, shares, impressions, reach, engagement_rate, post_timestamp, post_type, and platform where present. The README asks the project owner to verify the original dataset publisher, licence, and whether records are real, anonymized, or synthetic before making claims about provenance. These details are not independently established by the uploaded files.

4. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, and Google Colab/Jupyter Notebook.

5. Methodology

5.1 Data loading: The notebook checks for the CSV in the dataset folder or current working directory. In Google Colab, it can request a file upload if the file is not found.

5.2 Data cleaning: Column names are standardized. The post_timestamp field is parsed as a datetime, and metric columns are converted to numeric values. Exact duplicate rows are removed. Missing values are reported rather than automatically replaced with zero.

5.3 Feature engineering: The notebook derives total_engagement from likes, comments, and shares. It also derives post_date, post_month, and post_hour from the timestamp when available.

5.4 Metric definitions: Total Engagement = Likes + Comments + Shares. Engagement Rate (%) = (Total Engagement / Impressions) × 100. Rows with zero or missing impressions do not receive a calculated impressions-based rate.

5.5 Analysis: The notebook groups data to compare average engagement rate by post type, average engagement by posting hour, average impressions and reach by platform, and average monthly engagement.

6. Visualizations

The notebook defines four core static charts: (1) average engagement rate by post type, (2) average engagement by posting hour, (3) average impressions and reach by platform, and (4) average monthly engagement.

It also defines three interactive Plotly visualizations: a donut chart showing post-type distribution, an animated monthly engagement bar chart by post type, and an animated scatter plot comparing impressions with total engagement over time, with reach represented by bubble size and platform by color.

7. Results and Discussion

The uploaded notebook contains the code for the analysis and visualizations but does not contain executed cell outputs. The CSV dataset was not included in the upload. Therefore, numerical results, category rankings, strongest posting hours, platform comparisons, and month-to-month conclusions cannot be responsibly reported in this version.

After running the notebook with the dataset, the team should insert the actual summary tables and chart-supported observations here. In particular, distinguish averages from totals: the monthly chart is based on average total engagement by month.

8. Limitations

The analysis is limited to the supplied dataset and should not be generalized to all social media users or platforms. Posting-time patterns are descriptive and do not prove causation. Engagement-rate values depend on the denominator; this project calculates an impressions-based rate, not a reach-based or follower-based rate. Missing data and outliers may affect averages. Dataset provenance and redistribution licence require verification.

9. Conclusion

This project establishes a reproducible Python workflow for preparing social media post data, calculating engagement metrics, and exploring patterns by content type, platform, posting hour, and month. Its charts are intended to support descriptive interpretation rather than causal claims. The final conclusion should be completed after the notebook is executed with the source CSV and its actual outputs have been reviewed.

10. References and Acknowledgement

Project notebook: Social_Media_Engagement_Analysis.ipynb (uploaded project file).

Project README: README(2).md (uploaded project file).

Dataset source and licence: To be confirmed and cited by the project team using the original dataset publisher's information.